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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 03/19/2007 |
| **PI Name:** | Voorhees, Peter W. Ph.D. | | |
| **Project Title:** | Coarsening in Solid-Liquid Mixtures (CSLM)-2 | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Metals and alloys | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | p-voorhees@northwestern.edu | **Fax:** | FY 847-491-7820 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 847-491-7815 |
| **Organization Name:** | Northwestern University | | |
| **PI Address 1:** | Department for Materials Science and Engineering | | |
| **PI Address 2:** | 2220 Campus Dr | | |
| **PI Web Page:** | | | |
| **City:** | Evanston | **State:** | IL |
| **Zip Code:** | 60208-0893 | **Congressional District:** | 9 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 05/01/2000 | **End Date:** | 07/26/2007 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 3 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Duval, W. | **Contact Phone:** | 216-433-5023 |
| **Contact Email:** | walter.m.duval@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE (3/21/07): Project may be funded longer but current end date is 7/26/2007 (per N. Hall/ GRC)<br>NOTE: Project is still funded for two years, per PI (3/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG3-2474 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials. | | |

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| **Task Description:** | Previous NASA sponsored work clearly showed that solid-liquid mixtures consisting of Sn-rich particles in a Pb-Sn eutectic liquid are ideal, and perhaps unique, systems in which to explore the dynamics of the Ostwald ripening process. The high coarsening rate in these systems permit accurate kinetic data to be obtained, and the thermophysical parameters necessary to make a comparison between theory and experiment are known. However, in a terrestrial environment experiments can be performed only at the relatively high volume fractions of solid where the presence of a solid skeletal structure prevents large-scale particle sedimentation. In these high volume fraction experiments, a comparison between theory and experiment shows that the solid-liquid mixtures are coarsening faster than predicted by an approximate theory for purely diffusional controlled Ostwald ripening. The objective of this project is to plan and perform a microgravity experiment on Ostwald ripening in solid-liquid mixtures. This experiment will serve two primary purposes: it will allow experiments to be performed which can be directly compared to heretofore untested theories for coarsening in systems with low volume fractions of solid, and it will eliminate conclusively convection of the liquid matrix and small-scale particle motion within the skeletal structure as possible sources for the disagreement observed between theory and experiment in the high volume fraction experiments.<br><br>To perform coarsening experiments whose results can be compared directly with theory with no adjustable parameters.<br><br>The spaceflight experiment will produce data which, for the first time, can be compared directly to theory with no adjustable parameters. This data will address the long standing controversy over the dependence of the coarsening rate of a two-phase system on the volume fraction of coarsening phase. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | CSLM-2 will make possible predictions of coarsening kinetics in alloys with high volume fractions of precipitates, such as alloys under consideration for high temperature exploration applications (i.e., nuclear propulsion, waste heat coolant loops). Results will be applied to materials design codes that will reduce cost, shorten development time, and improve performance over conventional materials design methods. |
| **Task Progress:** | No progress report received this reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 01/17/2024) |